

Linking socioeconomic status to depressive symptoms in aging populations

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Late-life depressive symptoms remain a pressing global health issue, particularly among older adults with persistent socioeconomic disadvantages. While previous studies have linked socioeconomic status (SES) to mental health, few have integrated dynamic SES trajectories, psychosocial mechanisms and cultural contexts in a unified life course framework. Drawing on harmonized longitudinal data from 6 global aging cohorts (Health and Retirement Study (HRS), China Health and Retirement Longitudinal Study (CHARLS), Survey of Health, Ageing and Retirement in Europe (SHARE), English Longitudinal Study of Ageing (ELSA), Mexican Health and Aging Study (MHAS) and Korean Longitudinal Study of Aging (KLoSA); $n = 64,479$), this study examined how childhood SES, adult SES and SES mobility predicted incident depressive symptoms. A moderated chain mediation model tested the indirect effects of frailty and social activity, and the buffering role of family support. Cross-lagged panel network analysis was used to explore symptom-level directional associations. The results showed that higher childhood and adult SES, and stable high and upward SES trajectories, were consistently associated with lower risk of late-life depressive symptoms. Frailty and reduced social activity partially mediated these associations, while family support moderated the frailty–depressive symptoms pathway. Network analysis revealed temporal, symptom-level effects of adult SES on depressive symptoms, particularly in low- and middle-income countries. Subgroup analysis indicated stronger SES protection among men, with stable middle SES benefiting women more. Socioeconomic advantage across the lifespan protected against late-life depressive symptoms via social pathways, with culturally embedded family support acting as a key buffer. These findings highlight the importance of integrated, equity-focused and culturally sensitive strategies to sustain mental health in aging populations.

Late-life depressive symptoms are a significant global public health concern¹, affecting approximately 13.3% of older adults², with the highest burden observed among those aged 65–70 years³. Despite advances in pharmacological and psychological treatments, fewer than 40% of individuals with late-life depressive symptoms achieve long-term remission^{4,5}. This underscores the necessity to move beyond singular pathological perspectives and consider systemic and structural determinants of mental health^{6,7}.

Socioeconomic status (SES) is a critical determinant of mental health in later life. Existing theories highlight different pathways through which SES influences health^{8,9}. The social gradient in health concept posits that health outcomes systematically improve with higher SES and deteriorate with lower SES¹⁰. The cumulative disadvantage hypothesis suggests that adverse effects compound over time¹¹, whereas the sensitive period hypothesis contends that disadvantages during childhood can leave lasting ‘biological imprints’

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Table 1 | Baseline characteristics of participants in pooled cohorts

Variables	Total (n=64,479)	Non-depressive symptoms (n=39,663)	Depressive symptoms (n=24,816)	P
Survival time (years), mean (s.d.)	6.99 (4.94)	8.25 (5.22)	4.97 (3.64)	<0.001
Sex, n (%)				<0.001
Men	31,742 (49.23)	20,773 (52.37)	10,969 (44.20)	
Women	32,737 (50.77)	18,890 (47.63)	13,847 (55.80)	
Age (years), mean (s.d.)	64.97 (9.36)	64.80 (9.34)	65.26 (9.37)	<0.001
Marital status, n (%)				0.008
Other	14,527 (22.53)	8,798 (22.18)	5,729 (23.09)	
Married	49,952 (77.47)	30,865 (77.82)	19,087 (76.91)	
Parental survival status, n (%)				<0.001
Both dead	34,297 (53.19)	20,118 (50.72)	14,179 (57.14)	
One alive	12,083 (18.74)	7,694 (19.40)	4,389 (17.69)	
Both alive	18,099 (28.07)	11,851 (29.88)	6,248 (25.18)	
Smoking status, n (%)				<0.001
Never or ever smoker	51,766 (80.28)	32,071 (80.86)	19,695 (79.36)	
Current smoker	12,713 (19.72)	7,592 (19.14)	5,121 (20.64)	
Alcohol consumption, n (%)				<0.001
No	26,421 (40.98)	14,972 (37.75)	11,449 (46.14)	
Yes	38,058 (59.02)	24,691 (62.25)	13,367 (53.86)	
Exercise, n (%)				<0.001
No	36,511 (56.62)	20,980 (52.90)	15,531 (62.58)	
Yes	27,968 (43.38)	18,683 (47.10)	9,285 (37.42)	
Childhood SES, n (%)				<0.001
0 (low)	37,549 (58.23)	21,858 (55.11)	15,691 (63.23)	
1 (middle)	19,462 (30.18)	12,832 (32.35)	6,630 (26.72)	
2 (high)	7,468 (11.58)	4,973 (12.54)	2,495 (10.05)	
Adult SES, n (%)				<0.001
0 (low)	19,228 (29.82)	9,782 (24.66)	9,446 (38.06)	
1 (middle)	26,631 (41.30)	16,782 (42.31)	9,849 (39.69)	
2 (high)	18,620 (28.88)	13,099 (33.03)	5,521 (22.25)	
SES change, n (%)				<0.001
0 (downward)	10,433 (16.18)	5,890 (14.85)	4,543 (18.31)	
1 (stable low)	11,907 (18.47)	4,853 (12.24)	7,054 (28.43)	
2 (stable middle)	7,458 (11.57)	4,981 (12.56)	2,477 (9.98)	
3 (stable high)	2,465 (3.82)	1,890 (4.77)	575 (2.32)	
4 (upward)	32,216 (49.96)	22,049 (55.59)	10,167 (40.97)	

Survival time and age were described by mean (s.d.). The differences of categorical variables were assessed by χ^2 tests, and the differences of continuous variables were compared by t-tests. The two-sided test was used for statistical analysis. Scores for Childhood SES, Adult SES and SES change were coded from 0 to 4, with higher scores indicating higher socioeconomic status. For SES change, 0 = downward mobility, 1 = stable low, 2 = stable middle, 3 = stable high and 4 = upward mobility.

that persist despite subsequent SES improvements¹². In addition, the social mobility hypothesis emphasizes the role of SES changes across the life course¹³. Upward mobility may buffer the adverse effects of early life disadvantage, while downward mobility may increase stress and vulnerability to depressive symptoms. These theories help us to clarify the relative influence of different SES patterns on depressive symptoms in older adults¹⁴.

Building on this framework, we consider how SES may exert its influence on depressive symptoms through key intermediary processes. Previous research suggests that prolonged exposure to socioeconomic disadvantage can lead to both biological deterioration and reduced social integration—two core mechanisms implicated in the onset and persistence of depressive symptoms in later life^{15–18}. Accordingly, we propose that the effects of SES on late-life depressive

symptoms are transmitted through two primary pathways: physiological vulnerability and social disconnection. Specifically, frailty reflects the physiological consequences of cumulative disadvantage, while reduced social activity captures diminished social participation resulting from material constraints and role loss¹⁹. However, the strength of these indirect effects may vary depending on protective interpersonal resources. In many aging societies—particularly those with strong familial norms—intergenerational support plays a central role in sustaining psychological well-being²⁰. We therefore examine whether family support moderates the indirect pathways by buffering the psychological risks associated with frailty and social isolation. This psychosocial perspective enables us to distinguish between mediating and moderating mechanisms and to identify culturally relevant intervention targets that address both

structural and interpersonal determinants of mental health in aging populations^{21–23}.

Existing studies^{24,25} on SES and late-life depressive symptoms are limited by static SES indicators, isolated mediator analyses and single-country samples. To address these gaps, this study adopts a psychosocial model to elucidate how shifts in SES from childhood to old age contribute to depressive symptom risk²⁶. By integrating frailty, social activity and family support into a unified analytic framework and drawing on six major longitudinal aging cohorts (Health and Retirement Study (HRS), China Health and Retirement Longitudinal Study (CHARLS), Survey of Health, Ageing and Retirement in Europe (SHARE), English Longitudinal Study of Ageing (ELSA), Mexican Health and Aging Study (MHAS) and Korean Longitudinal Study of Aging (KLoSA)), we aim to identify culturally relevant intervention targets for precise, equity-focused prevention. Ultimately, this work seeks to shift the emphasis from fragmented, disease-centered approaches to holistic, structurally informed strategies for sustaining mental health in older populations. Beyond the social gradient perspective, our study design also enables empirical tests of other life course frameworks. By separately modeling childhood and adult SES, we address the critical period model, which posits that early life disadvantage leaves long-lasting biological imprints. By incorporating SES trajectories over time, our approach aligns with the cumulative advantage and disadvantage model, which emphasizes the compounding effects of long-term inequality. Finally, by explicitly distinguishing upward and downward mobility, we examine the social mobility hypothesis, which highlights how changes in SES across the lifespan shape health outcomes. This integrative framework provides a theoretically comprehensive foundation for the analyses that follow.

Results

Baseline characteristics of participants

Table 1 presents the baseline characteristics of all participants. Among the 64,479 respondents (average age 64.97 years, 50.77% women), a total of 24,816 individuals (38.49%) developed depressive symptoms during follow-up. The comparative analysis indicated that participants with depressive symptoms were more likely to be women, unmarried, both parents deceased, current smokers, non-drinkers and non-exercisers. In addition, those with depressive symptoms showed lower childhood SES and adult SES, and were more likely to show stable low SES or decline in SES over time. Similar patterns were observed across the six cohorts, with detailed results available in Supplementary Tables 8–13.

Association between SES status and depressive symptoms

The Cox proportional hazards model results suggested that higher childhood and adult SES were significantly associated with lower depressive symptoms. Compared with participants with downward SES, there was a significant negative correlation between stable high SES and depressive symptoms, followed by upward and stable middle SES, whereas stable low SES was positively correlated with depressive symptoms, regardless of covariate adjustments (Table 2). Taking model 3 of the pooled cohort as an example, compared with the lowest (downward) SES category, the hazard ratios (HRs) and 95% confidence intervals (CIs) were as follows: for middle childhood SES, 0.80 (95% CI, 0.77–0.82); high childhood SES, 0.70 (95% CI, 0.67–0.73); middle adult SES, 0.75 (95% CI, 0.73–0.77); high adult SES, 0.64 (95% CI, 0.61–0.66); stable low SES, 1.37 (95% CI, 1.32–1.43); stable middle SES, 0.76 (95% CI, 0.72–0.79); stable high SES, 0.54 (95% CI, 0.50–0.59) and upward SES, 0.69 (95% CI, 0.66–0.72). The same trend was observed in the HRS, CHARLS, SHARE, ELSA, MHAS and KLoSA cohorts, and in different country groups within the SHARE cohort (Supplementary Table 14). Overall, these findings indicated that higher childhood and adult SES and stable high SES mobility reduced the risk of developing depressive symptoms in later life. When

childhood and adult SES were adjusted for each other in the model, the results remained consistent (Supplementary Table 15). When we used Kaplan–Meier survival curves (Supplementary Figs. 6–8), analyzed SES by age category (Supplementary Table 16), restricted the analysis sample (Supplementary Table 17), used a competing risk model (Supplementary Table 18) or applied a mixed-effect cumulative logit model (Supplementary Fig. 9), the association between SES and depressive symptoms remained robust.

Moderated chain mediation analysis

A moderated chain mediation model based on the pooled cohort showed that frailty status and social activity mediated the effect of SES on depressive symptoms, suggesting that lower childhood SES, adult SES, and downward or stable low SES were positively correlated with depressive symptoms by exacerbating frailty status and reducing social activity (Fig. 1 and detailed data shown in Supplementary Table 19). Furthermore, family support moderated the impact of frailty status on depressive symptoms, indicating that individuals experiencing frailty who received financial or emotional support from family members were negatively correlated with depressive symptoms. These findings were consistently supported across the HRS, CHARLS, SHARE, ELSA, MHAS and KLoSA cohorts (Supplementary Figs. 10–15).

Cross-lagged panel network analysis

The cross-lagged panel network analysis was conducted using data from baseline (T1) and the first follow-up (T2) in each cohort. Results showed that adult SES indicators at T1 negatively predicted depressive symptom indicators at T2 overall (Fig. 2). In particular, in developing countries (CHARLS and MHAS cohorts), there were negative correlations between each index of adult SES and each index of depressive symptoms. The specific edge weights (regularized logistic regression coefficients) are shown in Supplementary Tables 20–25. In addition, we conducted stability tests for the network models, with the results indicating that the models showed strong robustness (Supplementary Figs. 16 and 17).

Sex subgroup analysis

In the sex-stratified subgroup analysis, the protective effects of higher childhood SES, adult SES and stable high SES against depressive symptoms were found to be more pronounced in men than in women. However, the harmful effect of stable low SES was also more obvious in men, whereas stable middle SES had a stronger protective effect in women (Fig. 3). Detailed analyses are presented in Supplementary Tables 26–28.

Discussion

Using longitudinal data from six global aging cohorts, this study provides compelling evidence that life course SES—particularly low childhood SES, low adult SES and downward SES mobility—increases the risk of late-life depressive symptoms. These effects are partially mediated by frailty and reduced social activity, reflecting both biological vulnerability and social disengagement. Importantly, family support buffers the impact of frailty on depressive symptoms, underscoring its protective role. Furthermore, cross-lagged panel network analysis shows that higher adult SES prospectively predicts lower depressive symptoms, especially in low- and middle-income countries, offering a dynamic view of associations between SES and depressive symptoms over time.

Our findings resonate not only with the social gradient in health theory but also with several complementary life course theories. The significant independent effect of childhood SES across cohorts provides support for the critical period model²⁷, underscoring how early life disadvantage exerts persistent influence on later mental health. The heightened risk associated with long-term low SES and downward mobility reflects the cumulative advantage and disadvantage model, in which disadvantages accumulate over time to amplify vulnerability²⁸. Meanwhile, the protective effect of upward SES mobility and the heightened risk of downward mobility directly corroborate the social

Table 2 | Specific and pooled effects of SES on depressive symptoms

Model	HRS (n=11,925)	CHARLS (n=6,077)	SHARE (n=28,385)	ELSA (n=6,002)	MHAS (n=6,727)	KLoSA (n=5,363)	Pooled (n=64,479)
	HR (95% CI)	P	HR (95% CI)	P	HR (95% CI)	P	HR (95% CI)
(a) Childhood SES							
Model 1							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.78 (0.72–0.83)	<0.001	0.81 (0.74–0.89)	<0.001	0.75 (0.66–0.84)	<0.001	0.78 (0.76–0.81)
High	0.63 (0.55–0.74)	<0.001	0.69 (0.64–0.75)	<0.001	0.65 (0.57–0.74)	<0.001	0.65 (0.62–0.69)
Model 2							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.77 (0.72–0.83)	<0.001	0.80 (0.74–0.88)	<0.001	0.73 (0.65–0.83)	<0.001	0.79 (0.77–0.82)
High	0.64 (0.55–0.74)	<0.001	0.71 (0.66–0.77)	<0.001	0.66 (0.58–0.75)	<0.001	0.69 (0.66–0.72)
Model 3							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.79 (0.74–0.85)	<0.001	0.81 (0.74–0.89)	<0.001	0.73 (0.65–0.82)	<0.001	0.80 (0.77–0.82)
High	0.66 (0.57–0.77)	<0.001	0.72 (0.66–0.78)	<0.001	0.66 (0.58–0.75)	<0.001	0.70 (0.67–0.73)
(b) Adult SES							
Model 1							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.59 (0.54–0.64)	<0.001	0.75 (0.70–0.80)	<0.001	0.77 (0.69–0.86)	<0.001	0.65 (0.60–0.71)
High	0.47 (0.44–0.51)	<0.001	0.49 (0.41–0.57)	<0.001	0.64 (0.57–0.71)	<0.001	0.49 (0.42–0.57)
Model 2							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.60 (0.55–0.65)	<0.001	0.76 (0.72–0.82)	<0.001	0.83 (0.78–0.87)	<0.001	0.82 (0.75–0.90)
High	0.50 (0.46–0.54)	<0.001	0.53 (0.45–0.62)	<0.001	0.75 (0.71–0.80)	<0.001	0.60 (0.51–0.70)
Model 3							
Low	1 (reference)		1 (reference)		1 (reference)		1 (reference)
Middle	0.63 (0.57–0.68)	<0.001	0.77 (0.72–0.82)	<0.001	0.83 (0.79–0.88)	<0.001	0.82 (0.75–0.90)
High	0.54 (0.50–0.59)	<0.001	0.53 (0.45–0.63)	<0.001	0.77 (0.73–0.82)	<0.001	0.60 (0.51–0.71)
(c) SES change							
Model 1							
0 (downward)	1 (reference)		1 (reference)		1 (reference)		1 (reference)
1 (stable low)	1.36 (1.22–1.52)	<0.001	1.42 (1.31–1.53)	<0.001	1.69 (1.50–1.90)	<0.001	1.23 (1.12–1.35)
2 (stable middle)	0.67 (0.61–0.75)	<0.001	0.83 (0.73–0.95)	<0.001	0.76 (0.68–0.85)	<0.001	0.66 (0.57–0.76)

Table 2 (continued) | Specific and pooled effects of SES on depressive symptoms

Model	HRS (n=11,925)			CHARLS (n=6,077)			SHARE (n=28,385)			ELSA (n=6,002)			MHAS (n=6,727)			KLoSA (n=5,363)			Pooled (n=64,479)		
	HR (95% CI)	P		HR (95% CI)	P		HR (95% CI)	P		HR (95% CI)	P		HR (95% CI)	P		HR (95% CI)	P		HR (95% CI)	P	
3 (stable high)	0.42 (0.35–0.52)	<0.001		0.55 (0.44–0.69)	<0.001		0.41 (0.29–0.57)	<0.001		0.56 (0.48–0.65)	<0.001		0.46 (0.34–0.63)	<0.001		0.47 (0.36–0.63)	<0.001		0.50 (0.45–0.54)	<0.001	
4 (upward)	0.58 (0.53–0.63)	<0.001		0.77 (0.71–0.84)	<0.001		0.73 (0.66–0.82)	<0.001		0.73 (0.66–0.82)	<0.001		0.71 (0.64–0.80)	<0.001		0.66 (0.59–0.75)	<0.001		0.66 (0.64–0.69)	<0.001	
Model 2																					
0 (downward)	1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)		
1 (stable low)	1.36 (1.22–1.52)	<0.001		1.40 (1.29–1.51)	<0.001		1.58 (1.41–1.78)	<0.001		1.63 (1.36–1.95)	<0.001		1.29 (1.17–1.42)	<0.001		1.11 (1.01–1.22)	0.025		1.37 (1.32–1.43)	<0.001	
2 (stable middle)	0.68 (0.61–0.76)	<0.001		0.83 (0.72–0.94)	0.004		0.77 (0.69–0.86)	<0.001		0.80 (0.71–0.91)	0.001		0.76 (0.63–0.92)	0.005		0.76 (0.65–0.88)	<0.001		0.75 (0.71–0.79)	<0.001	
3 (stable high)	0.45 (0.37–0.55)	<0.001		0.59 (0.47–0.74)	<0.001		0.44 (0.31–0.61)	<0.001		0.58 (0.50–0.67)	<0.001		0.49 (0.36–0.67)	<0.001		0.54 (0.40–0.71)	<0.001		0.53 (0.48–0.58)	<0.001	
4 (upward)	0.61 (0.55–0.67)	<0.001		0.79 (0.72–0.86)	<0.001		0.74 (0.66–0.83)	<0.001		0.75 (0.67–0.84)	<0.001		0.72 (0.65–0.81)	<0.001		0.73 (0.64–0.83)	<0.001		0.68 (0.65–0.71)	<0.001	
Model 3																					
0 (downward)	1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)			1 (reference)		
1 (stable low)	1.34 (1.20–1.49)	<0.001		1.39 (1.28–1.51)	<0.001		1.56 (1.39–1.76)	<0.001		1.61 (1.34–1.94)	<0.001		1.29 (1.17–1.42)	<0.001		1.12 (1.02–1.23)	0.021		1.37 (1.32–1.43)	<0.001	
2 (stable middle)	0.71 (0.64–0.79)	<0.001		0.83 (0.73–0.95)	0.006		0.77 (0.69–0.86)	<0.001		0.81 (0.72–0.93)	0.002		0.76 (0.63–0.92)	0.005		0.76 (0.65–0.88)	<0.001		0.76 (0.72–0.79)	<0.001	
3 (stable high)	0.49 (0.40–0.60)	<0.001		0.60 (0.48–0.75)	<0.001		0.45 (0.32–0.62)	<0.001		0.60 (0.52–0.69)	<0.001		0.49 (0.36–0.67)	<0.001		0.54 (0.40–0.72)	<0.001		0.54 (0.50–0.59)	<0.001	
4 (upward)	0.65 (0.59–0.71)	<0.001		0.79 (0.72–0.86)	<0.001		0.74 (0.67–0.83)	<0.001		0.77 (0.68–0.86)	<0.001		0.72 (0.65–0.81)	<0.001		0.73 (0.64–0.83)	<0.001		0.69 (0.66–0.72)	<0.001	

Model 1 did not make any adjustments. Model 2 adjusted sex, age, marital status and parental survival status. Model 3 further adjusted smoking, alcohol consumption and exercise on the basis of model 2. In particular, in the SHARE and pooled cohorts, we controlled for different countries. The two-sided test was used for statistical analysis and adjusted for multiple comparisons.

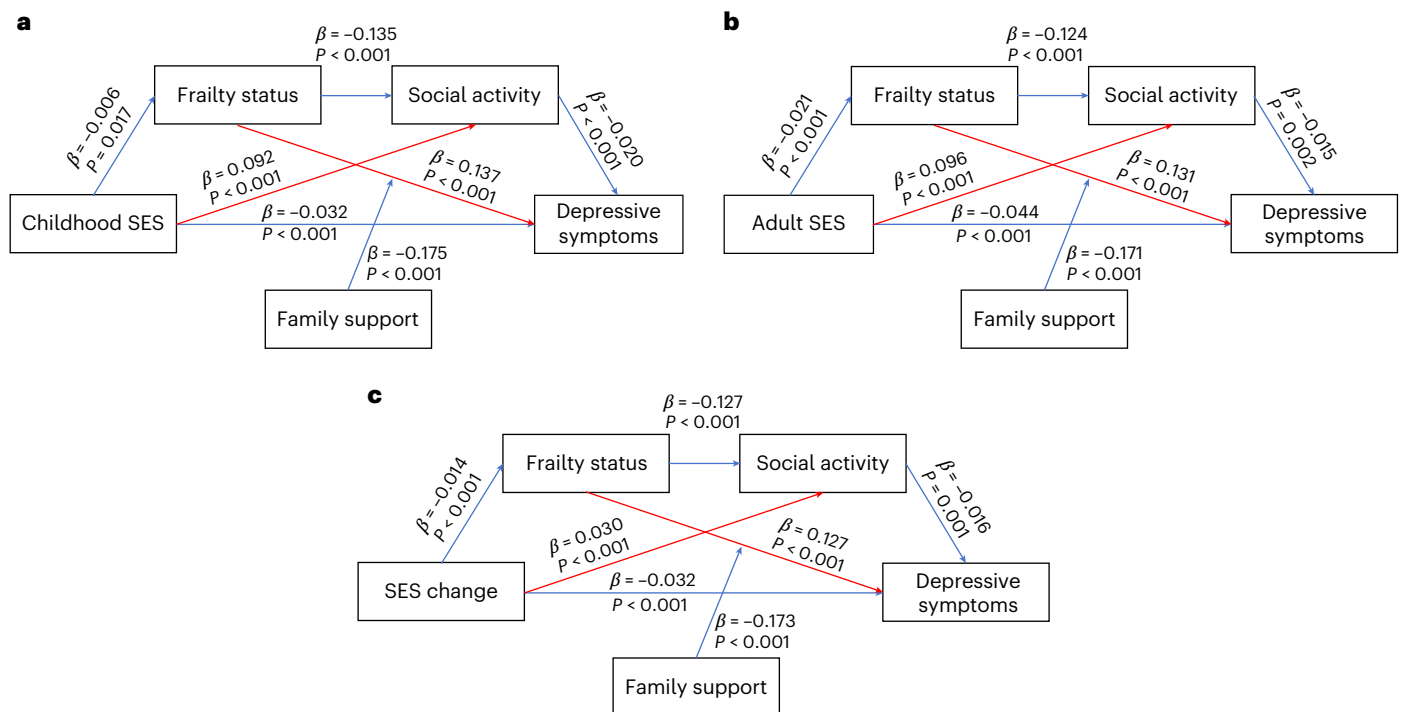


Fig. 1 | Chain mediating effects of frailty status and social activity on depressive symptoms, moderated by family support. a–c. Shown are the chain mediating effects of frailty status and social activity on depressive symptoms in relation to childhood SES (**a**), adult SES (**b**) and SES change (**c**), with family support as a moderator in the pooled cohort. The total analysis sample included 32,741 people (HRS, $n = 8,378$; CHARLS, $n = 2,356$; SHARE, $n = 11,624$; ELSA,

$n = 4,189$; MHAS, $n = 3,161$; KLoSA, $n = 3,033$). All analyses adjusted sex, age, marital status, parental survival status, smoking status, alcohol consumption, exercise and countries. The two-sided likelihood ratio test was used for statistical analysis and adjusted for multiple comparisons. β represents standardized path coefficients.

mobility framework. Together, these results highlight that multiple life course processes converge to shape late-life depression risk^{29,30}.

Our findings indicate that higher childhood SES, higher adult SES, and stable high or upward SES mobility are all associated with a reduced risk of later-life depressive symptoms^{23,31–33}. Specifically, the protective effects of both childhood and adult SES lend credence to the accumulation-sensitivity dual-pathway model embedded within the social gradient in health framework^{34,35}. Childhood SES influences neurodevelopmental plasticity through enhanced access to education and nutrition³⁶, whereas adult SES helps mitigate physiological decline through better healthcare access³³. In doing so, it challenges traditional static paradigms of SES measurement and signals that public health policies must simultaneously prioritize equitable opportunities for intergenerational mobility (for example, vocational training) and early life interventions (for example, child nutrition programs)^{37–39}.

From a mechanistic standpoint, low SES often correlates with factors such as malnutrition, insufficient chronic disease management and limited access to healthcare resources⁴⁰. These adverse conditions accelerate physiological aging and functional decline, evidenced by an elevated frailty index (FI)⁴¹. Among older adults, physiological capacity deteriorates naturally with advancing age, and those with a higher FI have diminished resilience to physical and psychological stress, making them more prone to depressive symptoms⁴². Concurrently, low SES constrains older adults' opportunities and abilities to participate in social activities⁴³. Older individuals facing economic hardship, limited transportation or scarce community resources engage in fewer daily social interactions and are less inclined to take part in community events and interpersonal networks⁴⁴. As social roles and connections progressively weaken, older adults become increasingly vulnerable to feelings of isolation and despair, which can trigger or exacerbate depressive symptoms⁴⁵. This study further elucidates a chain mediation mechanism by which SES affects depression. Low SES initially hastens

the depletion of physiological reserves through chronic stress^{26,46}, subsequently diminishing individuals' capacity to sustain social participation⁴⁷. Over time, these interconnected processes form a deteriorating pathway—from frailty to reduced social functioning and, ultimately, depressive symptoms. This finding provides additional support for the biopsychosocial model, underscoring the need for interventions that simultaneously address frailty management (for example, comprehensive geriatric assessments) and social reintegration (for example, establishing community activity centers)^{48,49}.

The family system constitutes a crucial protective factor for older adults' mental health⁵⁰. Our findings show that family support has a moderating effect on the frailty–depressive symptoms pathway, a result with significant practical implications. Previous research indicates that economic support from family can alleviate healthcare burdens, thereby reducing the release of stress hormones, while emotional support can facilitate oxytocin secretion, mitigating excessive amygdala activation⁵¹. The strength of this effect varies across countries, likely reflecting cultural differences in familial norms and intergenerational expectations. In East Asian societies, such as Korea and China, deeply rooted Confucian traditions emphasize filial piety and strong intergenerational bonds, which enhance the effectiveness of family support in mitigating the psychological burden associated with physical decline⁵². In contrast, while the United States is typically characterized by individualism, family support still plays an important role in later life⁵³, particularly among older adults experiencing health-related vulnerability or limited access to external resources. In addition, the presence of ethnic groups with strong familial networks—such as African American and Hispanic populations—may contribute to the protective role of family support in the US context⁵⁴. These findings highlight the need to consider cultural contexts when exploring psychosocial pathways of depressive symptoms and designing targeted mental health interventions for aging populations⁵⁵.

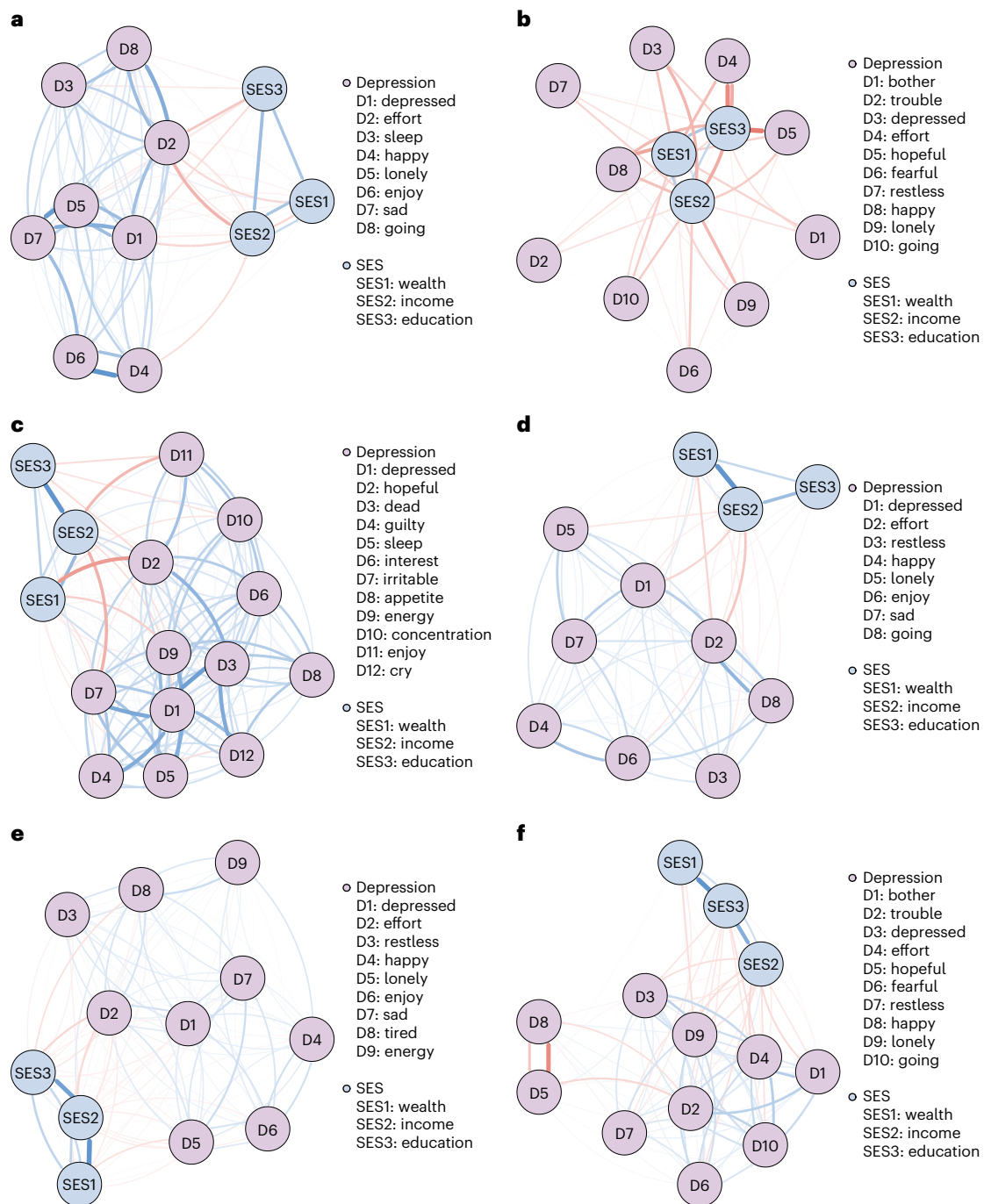


Fig. 2 | Network structures across HRS, CHARLS, SHARE, ELSA, MHAS and KLoSA. a–f. Network structures are shown for HRS ($n = 14,163$) (a), CHARLS ($n = 9,595$) (b), SHARE ($n = 33,041$) (c), ELSA ($n = 6,915$) (d), MHAS ($n = 9,543$) (e) and KLoSA ($n = 6,837$) (f). All analyses adjusted for sex, age, marital status, parental survival status, smoking status, alcohol consumption and exercise. In particular, we controlled for different countries in the SHARE cohort. The lines in the figure represent β , with values greater than 0 shown in blue and values less

than 0 shown in red. To enhance clarity, autoregressive effects and covariates are not shown in the figure. D1–D12 represent individual depressive symptoms derived from the CES-D scale. The numbering of D1–D12 varies between panels because the order of nodes is determined by the network estimation in each subgroup; therefore, the same code (for example, D8) may correspond to different symptoms across panels. SES1–SES3 represent the three socioeconomic status indicators: childhood SES, adult SES and SES change.

The cross-lagged panel network analysis further reinforces the temporal influence of SES on mental health outcomes. Specifically, adult SES indicators at baseline (T1) negatively predicted depressive symptom indicators at follow-up (T2), suggesting that higher socioeconomic resources may buffer the onset or exacerbation of depressive symptoms over time⁵⁶. This temporal pattern was particularly pronounced in developing countries such as China (CHARLS cohort) and Mexico (MHAS cohort), where each dimension of adult SES—including

wealth, income and education—show consistent inverse associations with subsequent depressive symptoms^{57,58}. These findings highlight the structural sensitivity of mental health to resource availability in lower-income contexts, where protective factors may be more unevenly distributed²³. By modeling symptom-level associations rather than aggregate scores, the network approach revealed specific pathways through which SES may influence depressive symptoms dynamically. Moreover, the robustness of the network structures across time points

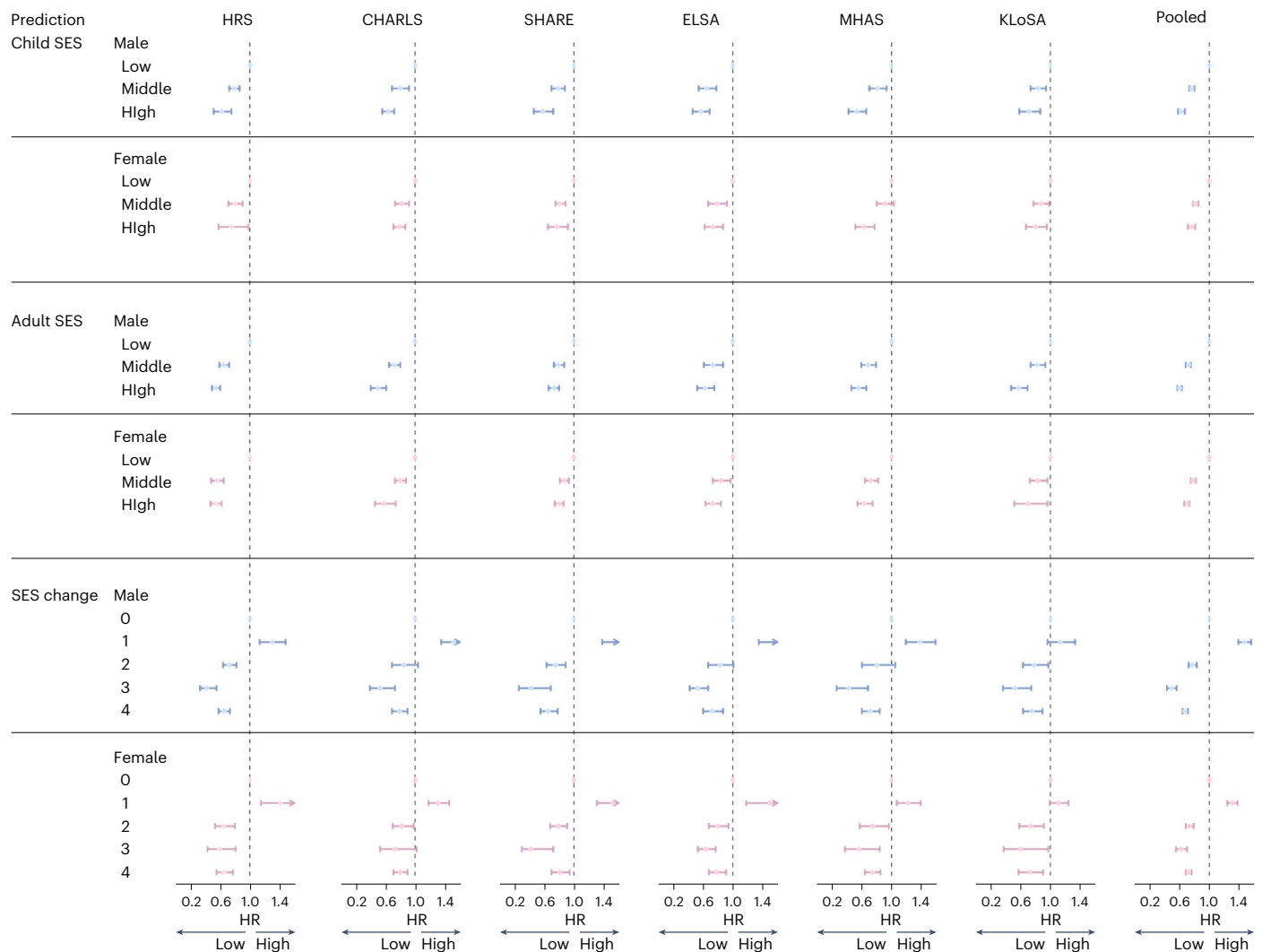


Fig. 3 | The effects of childhood SES, adult SES and SES change on depressive symptoms by sex. HRs are shown for each cohort: HRS ($n = 11,925$, 7,001 men and 4,924 women), CHARLS ($n = 6,077$, 3,104 men and 2,973 women), SHARE ($n = 28,385$, 14,879 men and 13,506 women), ELSA ($n = 6,002$, 3,228 men and 2,774 women), MHAS ($n = 6,727$, 3,769 men and 2,958 women) and KLoSA ($n = 5,363$, 2,833

men and 2,530 women). All analyses are adjusted for age, marital status, parent survival status, smoking status, alcohol consumption and exercise. In particular, we controlled for different countries in the SHARE and pooled cohorts. The points in the figure represent the HRs, and the lines represent the 95% CIs. Arrowheads indicate that the 95% confidence intervals extend beyond the plotted range.

underscores the potential utility of this method for tracking how socioeconomic vulnerabilities translate into psychological risk over time.

Sex-stratified analyses indicate that the protective effect of high SES level is stronger in men, while stable middle SES confers greater benefits for women. This sex-specific pattern may reflect differences in sex role expectations and stress sensitivity. Men are often expected to be the primary economic providers, making them more psychologically vulnerable to socioeconomic disadvantage or mobility⁵⁹. In contrast, women with stable middle SES may experience a balance between adequate material security and manageable social demands⁶⁰. This position may buffer against the dual pressures of career advancement and family caregiving often linked to middle or high SES, while still enabling access to essential social and emotional support—factors conducive to mental well-being in later life^{61,62}. However, these findings diverge from the predictions of the resource substitution theory, which posits that structurally disadvantaged groups—such as women—should derive greater psychological benefits from improved SES⁶³. Our results suggest that in midlife and older adulthood, women may benefit more from stability and moderation than from socioeconomic advancement⁶⁴. High SES in this population could carry added burdens, such as intergenerational caregiving

obligations and elevated role expectations, potentially undermining mental health⁶⁵. These findings point to a gendered nuance in how SES influences psychological outcomes in later life and suggest that resource-based theories may require adaptation when applied across the life course. Future research should investigate how gender roles and aging-related psychosocial dynamics jointly shape the mental health impacts of socioeconomic conditions.

This study makes several key contributions. Theoretically, it advances the life course perspective and reinforces the social gradient in health theory by showing how SES shapes depressive symptom risk through interconnected psychosocial processes, including functional vulnerability and reduced social engagement. Methodologically, it integrates latent class analysis, moderated chain mediation and cross-lagged panel network analysis to identify latent SES trajectories and dynamic symptom-level pathways over time. The network approach, in particular, captures directional interactions between SES indicators and depressive symptoms, offering richer insights than traditional regression models. Practically, the findings highlight the importance of fostering intergenerational socioeconomic mobility and strengthening family support systems as effective strategies for preventing late-life depressive symptoms.

Nonetheless, several limitations should be acknowledged. First, cross-cultural differences in variable measurement may affect the consistency and comparability of results. Second, retrospective self-reports of childhood SES are susceptible to recall bias and may not fully reflect actual early life conditions. Third, despite the advantages of temporal modeling, the lack of harmonized assessment tools across cohorts—particularly for SES, depressive symptoms and frailty—may introduce measurement bias and limit generalizability. In addition, controlling for unmeasured cultural, environmental and institutional confounders across countries remains a challenge. Finally, the influence of culture-specific family norms and social structures on depressive symptoms remains underexplored. Future research should adopt multisource, longitudinal and culturally adaptive designs and continue to leverage network-based and temporally sensitive methods to trace how structural disadvantage translates into psychological risk across the lifespan.

In conclusion, based on harmonized data from diverse aging cohorts, our research shows that higher childhood and adult SES and stable high and upward mobility are associated with lower risk of depressive symptoms in later life, while persistent disadvantage increases vulnerability via frailty and social disengagement. Family support serves as a key buffer, especially in cultures emphasizing filial piety. By applying cross-lagged panel network analysis, we highlight symptom-level, directional links between SES and depression over time. Taken together, these results provide empirical validation across four major life course frameworks—social gradient in health, critical period, cumulative advantage and disadvantage and social mobility frameworks—thereby reinforcing the theoretical grounding for equity-focused interventions. These findings offer globally relevant insights into culturally shaped psychosocial mechanisms and underscore the need for equitable, context-sensitive strategies to promote mental health in aging populations.

Methods

Study design and participants

This study used longitudinal data from 6 cohorts covering individuals aged 50 and above across 21 countries, specifically the HRS, CHARLS, SHARE, ELSA, MHAS and KLoSA cohorts (Supplementary Information, p. 4). These longitudinal surveys were designed using similar methodologies and their study samples are nationally representative. All cohorts included key indicators such as depressive symptoms, childhood and adult SES, frailty status, social activity and family support. To ensure comparability across cohorts, the waves analyzed for each survey were as follows: HRS (waves 5–16), CHARLS (waves 1–5), SHARE (waves 4–9), ELSA (waves 3–10), MHAS (waves 3–6) and KLoSA (waves 1–9).

Participants included in this study were aged 50 years or older and had reported information on childhood and adult SES and family support at baseline, along with incident depressive symptoms, frailty status and social activity in follow-up. In addition, they had undergone at least two consecutive assessments. We excluded participants who had depressive symptoms at baseline, those with missing data on depressive symptoms and relevant indicators, and those who were lost to follow-up. All participants were followed from the date of their baseline assessment until the earliest of the following events: the onset of depressive symptoms, the date of their last valid follow-up survey return or the end of the study period (Supplementary Fig. 1). The screening process for samples related to moderated chain mediation analyses and cross-lagged panel network analyses can be found in Supplementary Figs. 2 and 3. Ethical approval for all cohort surveys was obtained (Supplementary Information, p. 4).

Assessment of depressive symptoms

Each cohort used different scales to assess depressive symptoms: HRS and ELSA used the 8-item Center for Epidemiological Studies Depression Scale (CESD-8)⁶⁶, MHAS used the CESD-9 (ref. 67), CHARLS and KLoSA used the CESD-10 (ref. 68), and SHARE adopted the 12-item

Europe Depression Scale (EURO-D)⁶⁹. All these scales have been validated as effective tools for predicting depressive symptoms and have been widely tested in older adult populations. Following previous studies, we defined the diagnostic cutoff values for depressive symptoms in each cohort as follows: for HRS and ELSA, ≥ 4 ; for SHARE, ≥ 4 ; for MHAS, ≥ 5 ; for CHARLS and KLoSA, ≥ 10 (refs. 23,70). Further details can be found in Supplementary Table 1.

Assessment of SES

Previous studies have commonly measured SES using employment status, family income and education^{71,72}. For childhood SES, we assessed this construct using three indicators: whether the participant's father experienced unemployment during childhood, the degree of economic hardship in the family during childhood and parental education levels⁷³. However, due to a lack of relevant data in the KLoSA cohort, childhood SES in KLoSA was measured solely by parental education levels⁷⁴. Further details are provided in Supplementary Table 2. For adult SES, considering that the participants in the study were mostly retirees, we used family wealth instead of employment status⁷⁵. That is, we evaluated adult SES using family wealth, family income and educational attainment⁷⁶, with detailed descriptions available in Supplementary Table 3. To construct a composite index reflecting childhood and adult SES, we applied latent class analysis and identified three SES categories: low, middle and high (Supplementary Table 4). The specific probability distributions are shown in Supplementary Figs. 4 and 5. For SES change, we computed the difference between adult SES and childhood SES and classified it into 5 categories: 0 (downward), 1 (stable low), 2 (stable middle), 3 (stable high) and 4 (upward).

Assessment of mediating and moderating variables

Frailty status was assessed using the FI, which was derived from the accumulation of multiple age-related indicators, including chronic diseases, self-reported health, functional disability and cognitive function^{77,78}. Except for cognitive function, all other items were dichotomous, with 1 indicating the presence of a deficit and 0 indicating its absence. Cognitive function was treated as a continuous variable, ranging from 0 to 1, where higher scores indicate poorer cognitive ability. In this study, we defined frailty as FI ≥ 0.25 and coded it as 1 (ref. 79). Details of the FI components are provided in Supplementary Table 5. Social activity was measured based on participants' engagement in social group activities and categorized into two groups according to whether they participated in at least one social activity (no or yes). The specific types of activity are listed in Supplementary Table 6. Family support was defined as the extent to which family members provided financial assistance or to what extent they could be relied upon, and this was categorized as no versus yes. The measurement details are provided in Supplementary Table 7. To accurately measure the influence paths among various variables, frailty status was measured using the second-period data (non-frail in the first period), social activity was measured using the third-period data and family support was measured using the baseline data.

Covariates

On the basis of existing studies, this study identified demographic characteristics—sex (men, women), age, marital status (other, married) and parent survival status (both dead, one alive or both alive)—and lifestyle factors—smoking status (never or ever smoker, current), alcohol consumption (no, yes), and exercise (no, yes)—as potential confounding factors and covariates. Further details are provided in Supplementary Table 7.

Statistical analysis

In this study, baseline characteristics were described according to whether participants had depressive symptoms, using mean (s.d.) for continuous variables and frequency (percentage) for categorical variables. Differences were analyzed separately for each of the

six cohorts and for the pooled dataset. Furthermore, we also conducted comparative analyses of the samples included in the survival analysis and the excluded samples to illustrate the sample differences (Supplementary Tables 29–34).

To examine the impact of childhood SES, adult SES and SES change on depressive symptoms, we used a Cox proportional hazards model. Survival time was measured from baseline assessment to the first report of depressive symptoms, last valid follow-up or the end of the study. HRs with 95% CIs were estimated to assess the associations of interest. To ensure the robustness of findings, we adjusted for different covariates across models. Model 1 was unadjusted. Model 2 was adjusted for demographic characteristics (sex, age, marital status and parent survival status). Model 3 further adjusted for lifestyle factors, including smoking, alcohol consumption and exercise.

A moderated chain mediation model was used to analyze the mediating roles of frailty status and social activity in the associations of childhood SES, adult SES and SES change with depressive symptoms, and the moderating role of family support. In addition, to better capture the dynamic correlations between each index of depressive symptoms and each index of adult SES, we used cross-lagged panel network analysis to further explore the relationships among the indicators. A sex-stratified subgroup analysis was conducted to explore whether the association between SES and depressive symptoms differed between men and women.

To ensure reliability of the results, we performed five sensitivity analyses to strengthen the robustness of our findings. First, we used Kaplan–Meier survival curves to analyze differences in incident depressive symptoms across SES groups. Second, we treated age as a categorical variable (50–59 defined as 0, 60 and above defined as 1) and included it as a covariate. Third, we restricted the sample to individuals who were followed from baseline to the study end point, except for those excluded due to the onset of depressive symptoms. Fourth, we treated death as a competing risk for depressive symptoms and reanalyzed the data using a competing risk model. Finally, we applied a mixed-effects cumulative logit model to assess the relationship between SES and depressive symptoms, ensuring that the results for the pooled cohort remained robust despite unbalanced observations (that is, not all individuals were observed at the same time points). All analyses were conducted using STATA (v.16.0) and R (v.4.4.2), with a two-sided $P < 0.05$ considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The original data of each cohort can be obtained upon application on their respective official websites: HRS (<https://hrs.isr.umich.edu/data-products>), CHARLS (<https://charls.charlsdata.com/pages/data/111/en.html>), SHARE (<https://share-eric.eu>), ELSA (<https://beta.ukdataservice.ac.uk/datacatalogue/series/series?id=200011>), MHAS (<https://www.mhasweb.org/DataProducts/Home.aspx>) and KLoSA (<http://survey.keis.or.kr/eng/klosa/klosa01.jsp>). All the data used in this article have been uploaded to Zenodo at: <https://zenodo.org/records/17206621> (ref. 80).

Code availability

All data analyses used open-source STATA (v.16.0) and R (v.4.4.2) codes, and the relevant codes are publicly available. All the code involved in this analysis has been uploaded to Zenodo at: <https://zenodo.org/records/17206818> (ref. 81).

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Author contributions

Z.Z. and H.X. were responsible for data processing and writing the paper. Y.Y. collected and analyzed the data and contributed to the final paper. J.Z. and X.C. were responsible for proofreading, reviewing and securing funding.

Competing interests

The authors declare no competing interests.

Additional information

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Reporting on sex and gender	Sex was recorded as male or female in all datasets. Sex-stratified subgroup analyses were conducted.
Reporting on race, ethnicity, or other socially relevant groupings	Considering the scientific nature of the research, the classification of race or ethnicity is not involved in the analysis.
Population characteristics	Age, marital status, parent alive, smoke, drink, and exercise detailed in Table 1.
Recruitment	Participants were recruited by the original cohort study protocols, which followed rigorous population sampling strategies.
Ethics oversight	All data used in this study received ethical approval from the original survey institutions: HRS (approved by the National Institute on Aging and the Social Security Administration, NIA U01AG009740), CHARLS (approved by the Ethical Review Committee of Peking University, IRB00001052-11015), SHARE (approved by the Ethics Council of the Max Planck Society), ELSA (approved by the National Research Ethics Service Committee South Central-Berkshire), MHAS (approved by the National Institutes of Health/National Institute on Aging, NIH R01AG018016), and KLoSA (approval by the National Statistical Office). No additional approvals were required. All participants provided written informed consent.

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Study description	This study is a quantitative secondary analysis of existing data from six international longitudinal cohorts on aging (HRS, CHARLS, SHARE, ELSA, MHAS, KLoSA). No new data were collected. Statistical analyses were conducted to examine associations between socioeconomic status, and depressive symptoms outcome.
Research sample	We used data from five nationally or regionally representative cohorts (HRS, CHARLS, SHARE, ELSA, MHAS, and KLoSA), including participants aged 50 years and older. After exclusions, the final sample sizes were 11,925 (HRS), 6,077 (CHARLS), 28,385 (SHARE), 6,002 (ELSA), 6,727 (MHAS), and 5,363 (KLoSA). Demographic information such as age and sex was collected through self-report. These cohorts were chosen for their large, comparable samples and comprehensive measures of socioeconomic status, and depressive symptoms. For data sources, please refer to "Data" in the Reporting Summary or Data Availability in the paper.
Sampling strategy	Participants are selected based on inclusion criteria of age 50 or older, complete data on socioeconomic status, depressive symptoms, and covariates, and no depressive symptoms diagnosis at baseline. No probabilistic sampling was conducted by the authors; data were drawn from pre-established, nationally representative cohorts.
Data collection	This study was a secondary analysis of existing data from six longitudinal cohorts (HRS, CHARLS, SHARE, ELSA, MHAS, KLoSA). Data collection in these cohorts was conducted by trained personnel using standardized questionnaires and health assessments. Instruments included validated scales such as the Socioeconomic Status (SES), CES-D, and Euro-D. Data were self-reported by participants or measured according to standardized protocols. No researchers involved in this study were present during the original data collection. Since this was a secondary data analysis, blinding was not applicable.
Timing	Data were collected from six international longitudinal cohorts over the following periods: HRS (2000–2022), CHARLS (2011–2020), SHARE (2011–2021), ELSA (2006–2020), MHAS (2012–2021), and KLoSA (2006–2022). Each cohort conducted multiple waves of data collection within these periods without major interruptions.
Data exclusions	Data were excluded based on pre-established criteria: participants younger than 50 years, with missing data on baseline socioeconomic status, covariates, or follow-up depressive symptoms, and those diagnosed with depressive symptoms at baseline. The final analytic samples included 11,925 from HRS, 6,077 from CHARLS, 28,385 from SHARE, 6,002 from ELSA, 6,727 from MHAS, and 5,363 from KLoSA, as detailed in Supplementary Figure S1.

Non-participation

Non-participation (e.g., drop-out or lost to follow-up) is addressed in sensitivity analyses and was not directly handled by the authors, as this is secondary data analysis. Attrition is reported in cohort documents and included in robustness tests.

Randomization

This is a non-interventional, observational study using secondary data. No random assignment was applied.

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